MACIERZE

$A = [a_{ij}]$ $\qquad A_{n\times m}$

i → nr. w. $\qquad$ n → liczba wierszy

j → nr. k. $\qquad$ m → liczba kolumn

$$A = \begin{bmatrix}1&2&3\\-1&0&1\end{bmatrix}_{2\times3} \qquad B = \begin{bmatrix}-1&-2&1\\-1&1&2\end{bmatrix}_{2\times3}$$

$$A+B = \begin{bmatrix}0&0&4\\-2&1&3\end{bmatrix}_{2\times3}$$

1) $A_{n\times m} + B_{n\times m} = C_{n\times m}$

2) $A_{n\times m} \cdot B_{m\times k} = C_{n\times k}$

$A \cdot B \neq B \cdot A$

3) $(A_{n\times m})^T = A^T_{m\times n}$

$$B^T = \begin{bmatrix}-1&-1\\-2&1\\1&2\end{bmatrix}$$

$$A \cdot B^T = \begin{bmatrix}1&2&3\\-1&0&1\end{bmatrix}_{3\times2} \cdot \begin{bmatrix}-1&-1\\-2&1\\1&2\end{bmatrix}_{2\times3} = \begin{bmatrix}-1-4+3 & -1+2+6\\1+0+1 & 1+0+2\end{bmatrix} =$$

$$= \begin{bmatrix}-2&7\\2&3\end{bmatrix}_{2\times2}$$

$$A = \begin{bmatrix}1&0&2\\3&5&1\end{bmatrix} \qquad B = \begin{bmatrix}1&3\\7&5\\0&2\end{bmatrix} \qquad C = \begin{bmatrix}-1&2&0\\0&0&5\end{bmatrix}$$

$$A \cdot B = \begin{bmatrix}1&7\\38&36\end{bmatrix} \qquad B \cdot A = \begin{bmatrix}10&15&5\\22&25&19\\6&10&2\end{bmatrix}$$

A \ B	1 7 0	3 5 2
1 0 2	1	7
3 5 1	38	36

B \ A	1 3	0 5	2 1
1 3	10	15	5
7 5	22	25	19
0 2	6	10	2

$$2A + C = \begin{bmatrix}2&0&4\\6&10&2\end{bmatrix} + \begin{bmatrix}-1&2&0\\0&0&5\end{bmatrix} = \begin{bmatrix}1&2&4\\6&10&7\end{bmatrix}$$

$$A - C = \begin{bmatrix}1&0&2\\3&5&1\end{bmatrix} + \begin{bmatrix}1&-2&0\\0&0&-5\end{bmatrix} = \begin{bmatrix}2&-2&2\\3&5&-4\end{bmatrix}$$

A^T \ C	-1 0	2 0	0 5
1 3	2	2	15
0 5	0	0	25
2 1	-2	4	5

$$A^T \cdot C = \begin{bmatrix}1&3\\0&5\\2&1\end{bmatrix} \cdot \begin{bmatrix}-1&2&0\\0&0&5\end{bmatrix} =$$

$$A\cdot(B\cdot C)=\begin{bmatrix}6&-3\\0&-3\\-1&-1\end{bmatrix}$$

$$\begin{array}{cc|cc} & C & -1 & 2\\ B & & 1 & 1\\ \hline -1 & 2 & 3 & 0\\ -1 & 1 & 2 & -1\\ 0 & -1 & -1 & -1\end{array}
\qquad
\begin{array}{ccc|cc} & & B\cdot C & 3 & 0\\ & & & 2 & -1\\ A & & & -1 & -1\\ \hline 1 & 2 & 1 & 6 & -3\\ 0 & 1 & 2 & 0 & -3\\ 0 & 0 & 1 & -1 & -1\end{array}$$

$$(A\cdot B)\cdot C=\begin{bmatrix}6&-3\\0&-3\\-1&-1\end{bmatrix}$$

$$\begin{array}{ccc|cc} & & B & -1 & 2\\ & & & -1 & 1\\ A & & & 0 & -1\\ \hline 1 & 2 & 1 & -3 & 3\\ 0 & 1 & 2 & -1 & -1\\ 0 & 0 & 1 & 0 & -1\end{array}
\qquad
\begin{array}{cc|cc} & C & -1 & 2\\ A\cdot B & & 1 & 1\\ \hline -3 & 3 & 6 & -3\\ -1 & -1 & 0 & -3\\ 0 & -1 & -1 & -1\end{array}$$

Zad. 3

A. $\begin{vmatrix} j & j+1\\ 2 & 3\end{vmatrix}=A$

$$\det A=3j-(j+1)\cdot 2=3j-2j-2=j-2$$

B. $\begin{vmatrix} j+1 & 2\\ 4 & j-1\end{vmatrix}=A$

$$\det A=(j+1)(j-1)-8=j^2-1-8=-10$$

C. $\begin{vmatrix} -3 & 2 & 1\\ 2 & 0 & 2\\ -1 & -2 & 1\end{vmatrix}$

$$\det\begin{vmatrix} -3 & 2 & 1\\ 2 & 0 & 2\\ -1 & -2 & 1\end{vmatrix}=-3-4-4+1-12-4=-16-14=-30$$

$$\begin{matrix} -3 & 2 & 1\\ 2 & 0 & 2\end{matrix}$$

Zad. 5

$$\begin{array}{c|ccc} A \backslash B & \begin{matrix}1\\0\\0\end{matrix} & \begin{matrix}0\\0\\-3\end{matrix} & \begin{matrix}1\\2\\0\end{matrix} \\ \hline 3\ 2\ 1 & 3 & -3 & 7 \\ 2\ 1\ 0 & 2 & 0 & 4 \\ 1\ 0\ 0 & 1 & 0 & 1 \end{array}$$

$$\begin{vmatrix} 3 & -3 & 7 \\ 2 & 0 & 4 \\ 1 & 0 & 1 \end{vmatrix} = 0+0-12+0+0+6=-6$$
$$\begin{matrix} 3 & -3 & 7 \\ 2 & 0 & 4 \end{matrix}$$

$$\begin{vmatrix} 3 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 0+0+0-1-0-0=-1$$
$$\begin{matrix} 3 & 2 & 1 \\ 2 & 1 & 0 \end{matrix}$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & -3 & 0 \end{vmatrix} = 0+0+0-0+6+0=6$$
$$\begin{matrix} 1 & 0 & 1 \\ 0 & 0 & 2 \end{matrix}$$

$\det A \cdot \det B = -6$

$\det(A \cdot B) = \det A \cdot \det B$

c.n.w.

Zad. 6

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix} \qquad A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$